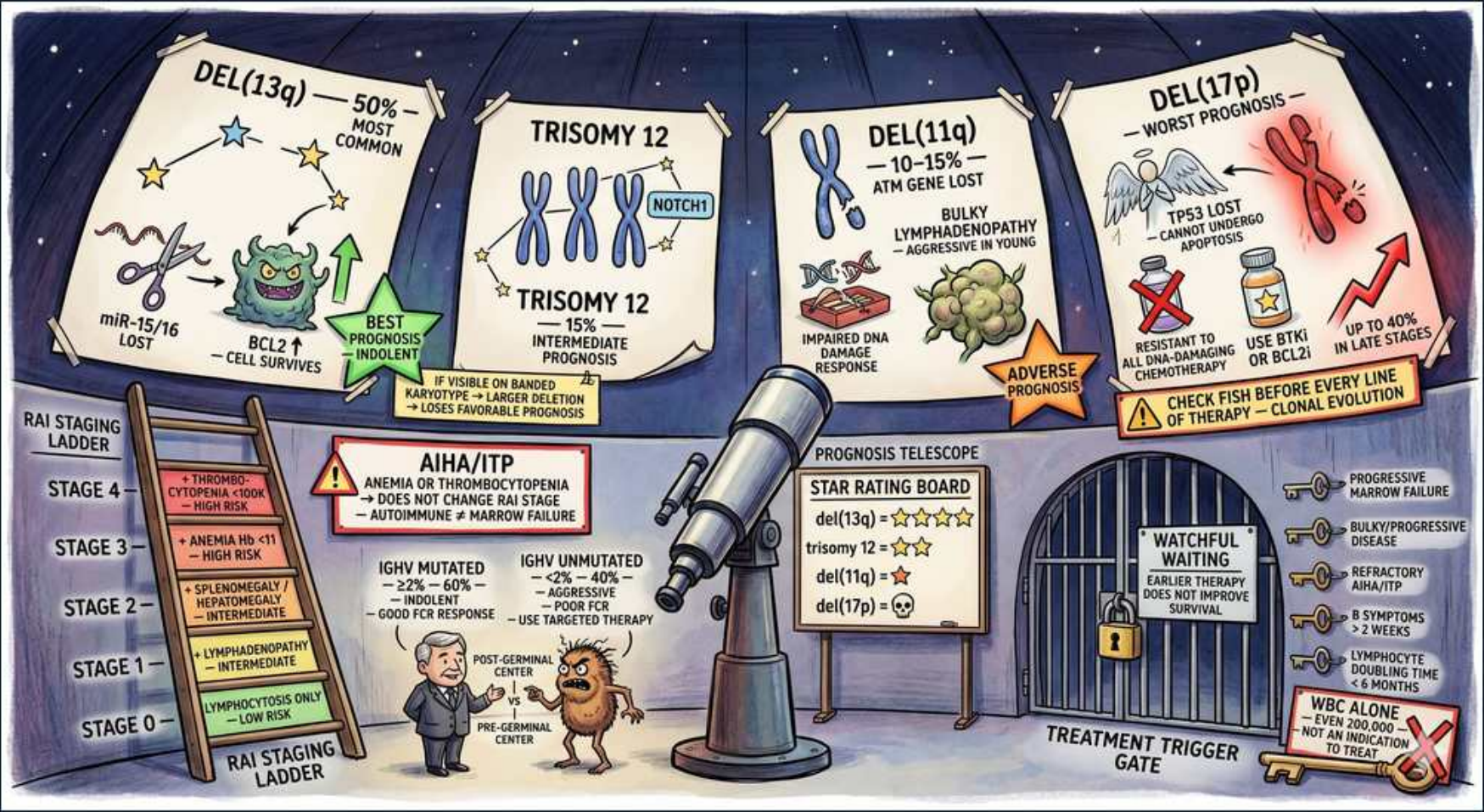


CLL — Prognosis & Treatment



1. del(13q) best

WHAT YOU SEE

A 'DEL(13q)' banner '50% MOST COMMON' showing scissors cutting miR-15/16 and a content green BCL2 monster surviving, green 'BEST PROGNOSIS — INDOLENT' arrow.

WHAT IT ENCODES

del(13q14) (loss of miR-15a/16-1 → unchecked BCL2 → cell survival) is the MOST COMMON CLL abnormality (~50%) and, when isolated, carries the BEST/most indolent prognosis with the longest survival.

BOARD TRAP

Best prognosis only when del(13q) is the SOLE abnormality — its favorable weight is overridden if del(17p)/TP53 or a complex karyotype coexists.

2. Trisomy 12

WHAT YOU SEE

A 'TRISOMY 12' banner with three blue X chromosomes and a NOTCH1 tag, '15% INTERMEDIATE PROGNOSIS'.

WHAT IT ENCODES

Trisomy 12 (~15%) confers an INTERMEDIATE prognosis and is associated with atypical morphology/immunophenotype and NOTCH1 mutations.

3. del(11q) adverse

WHAT YOU SEE

A 'DEL(11q)' banner '10-15% ATM gene lost' with shattered DNA and a bulky cluster of grapes (bulky lymphadenopathy), orange 'ADVERSE PROGNOSIS' tag.

WHAT IT ENCODES

del(11q22-23) deletes the ATM gene (impaired DNA-damage repair), producing BULKY lymphadenopathy and aggressive disease, often in younger patients — an ADVERSE marker. BTK inhibitors notably improve outcomes in del(11q) disease.

BOARD TRAP

del(11q) = bulky nodes + aggressive course; remember ATM is the lost gene, not TP53.

4. del(17p) worst

WHAT YOU SEE

A 'DEL(17p)' banner with a chromosome losing its winged TP53 and a red explosion, 'WORST PROGNOSIS'.

WHAT IT ENCODES

del(17p) deletes TP53 — the WORST prognosis, with resistance to DNA-damaging chemoimmunotherapy. Always test by FISH (and add TP53 sequencing, since TP53 mutation without 17p deletion is equally adverse).

BOARD TRAP

del(17p) AND/OR TP53 mutation both confer the worst prognosis — a normal FISH for 17p does not exclude a high-risk TP53 mutation; sequence TP53 too.

5. TP53 → no apoptosis

WHAT YOU SEE

'TP53 LOST — CANNOT UNDERGO APOPTOSIS' under the del(17p) banner — the p53 angel gone, so damaged cells refuse to die.

WHAT IT ENCODES

Loss of p53 means cells cannot trigger apoptosis after DNA damage, so they survive cytotoxic chemotherapy — the mechanistic reason del(17p)/TP53 CLL is chemoresistant and MUST be treated with targeted agents.

6. FISH before therapy

WHAT YOU SEE

A warning sign 'CHECK FISH BEFORE EVERY LINE OF THERAPY — CLONAL EVOLUTION' — re-test genetics each line because the clone shifts.

WHAT IT ENCODES

Repeat FISH (and TP53 sequencing) BEFORE EVERY line of therapy because clonal evolution can acquire del(17p)/TP53 over time, which changes drug selection (mandates BTKi/venetoclax over chemoimmunotherapy).

BOARD TRAP

A favorable FISH at diagnosis can evolve — never reuse old cytogenetics to plan a new line of treatment.

7. BTK / BCL2 for 17p

WHAT YOU SEE

Crossed-out chemo beside pill bottles 'USE BTKi OR BCL2i' — del(17p) is resistant to DNA-damaging chemo, so go targeted.

WHAT IT ENCODES

del(17p)/TP53 disease: use BTK inhibitors (ibrutinib/acalabrutinib/zanubrutinib) or the BCL2 inhibitor venetoclax — NOT chemoimmunotherapy (FCR/BR), which fails p53-deficient cells.

BOARD TRAP

Never give chemoimmunotherapy to del(17p)/TP53 CLL — it is ineffective and these patients require targeted therapy regardless of fitness.

8. IGHV mutated good

WHAT YOU SEE

A scholarly 'POST-GERMINAL CENTER' figure 'IGHV MUTATED 22-60% INDOLENT — good FCR response' — counterintuitively, mutated = favorable.

WHAT IT ENCODES

IGHV-MUTATED CLL (≥2% deviation from germline) is INDOLENT with favorable prognosis and durable responses; this is the subset that can achieve long remissions with fixed-duration chemoimmunotherapy (e.g., FCR) in TP53-intact, young, fit patients.

BOARD TRAP

"Mutated" sounds bad but is GOOD in CLL — mutated IGHV = favorable. (Counterintuitive: the opposite of the usual oncology connotation.)

9. IGHV unmutated poor

WHAT YOU SEE

An angry 'PRE-GERMINAL CENTER' monster 'IGHV UNMUTATED 2-40% AGGRESSIVE — poor FCR, use targeted' — unmutated = worse.

WHAT IT ENCODES

IGHV-UNMUTATED CLL (~40%) is AGGRESSIVE with shorter time-to-treatment and poorer survival, and responds poorly/short-lived to chemoimmunotherapy — favor continuous BTKi or venetoclax-based therapy.

BOARD TRAP

Unmutated IGHV = worse outcome; unlike most cancers, fewer mutations here means worse prognosis. IGHV status does not change over the disease course (test once).

10. AIHA / ITP

WHAT YOU SEE

A red 'AIHA/ITP' alert: 'anemia or thrombocytopenia — DOES NOT change Rai stage — autoimmune ≠ marrow failure'.

WHAT IT ENCODES

CLL causes AUTOIMMUNE cytopenias (warm AIHA, ITP) from immune dysregulation, not marrow replacement — treat with corticosteroids ± rituximab/IVIG, separate from CLL-directed therapy.

BOARD TRAP

Autoimmune anemia/thrombocytopenia does NOT upstage Rai — these are immune-mediated, distinct from the marrow-failure cytopenias that define Rai III/IV.

11. Rai staging ladder

WHAT YOU SEE

A 'RAI STAGING LADDER' climbing Stage 0 lymphocytosis (low) → 1 nodes → 2 spleen/liver → 3 anemia → 4 thrombocytopenia (high risk).

WHAT IT ENCODES

Rai staging ladder: 0 = lymphocytosis only (low risk); I = + lymphadenopathy; II = + splenomegaly/hepatomegaly (intermediate); III = + anemia (Hb <11); IV = + thrombocytopenia (plt <100) (high risk). Binet (A/B/C) is the European equivalent based on number of involved areas + cytopenias.

BOARD TRAP

Stage III/IV require MARROW-FAILURE cytopenias — anemia/thrombocytopenia from AUTOIMMUNITY does not count toward advanced stage.

12. Watch and wait

WHAT YOU SEE

A padlocked jail cell 'WATCHFUL WAITING — earlier therapy does NOT improve survival' — asymptomatic disease is locked away under observation.

WHAT IT ENCODES

Early/asymptomatic CLL (Rai 0–II without triggers) is OBSERVED ("watch and wait") — multiple trials show treating early-stage asymptomatic disease provides NO survival benefit and only adds toxicity.

BOARD TRAP

Do not treat asymptomatic early-stage CLL just because the diagnosis is made — observation is the standard of care.

13. Treatment triggers

WHAT YOU SEE

A 'TREATMENT TRIGGER GATE' with golden keys listing progressive marrow failure, bulky/progressive nodes, refractory AIHA/ITP, B symptoms, doubling <6 mo.

WHAT IT ENCODES

Treat only when iwCLL indications are met: progressive marrow failure (worsening anemia/thrombocytopenia), massive/progressive/symptomatic splenomegaly or lymphadenopathy, constitutional B symptoms, symptomatic disease, autoimmune cytopenia refractory to steroids, or lymphocyte doubling time <6 months.

BOARD TRAP

A high WBC/lymphocyte count ALONE is NOT an indication to treat — even very high counts without symptoms or end-organ effects don't warrant therapy (CLL rarely causes leukostasis).

14. Richter / clonal evolution

WHAT YOU SEE

The 'PROGNOSIS TELESCOPE' and star-rating board watching for clonal evolution — recheck FISH as the clone can acquire TP53 / transform to Richter.

WHAT IT ENCODES

CLL can transform (Richter) into aggressive DLBCL — suspect with rapid asymmetric nodal growth, B symptoms, and markedly elevated LDH; confirm with PET-guided excisional biopsy. Clonal evolution (esp. acquiring TP53) also drives resistance, so recheck genetics over time.

BOARD TRAP

Sudden clinical acceleration in known CLL = look for Richter (biopsy the PET-hot node), not merely CLL progression.
